CSA1717-ARTIFICIAL INTELLIGENCE

PRACTICAL LAB EXPERIMENTS

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Aim:

To write a Prolog program to implement **pattern matching** between two lists.

Objective:

To check whether a given pattern matches a sequence of elements using Prolog's **unification and recursion**.

Algorithm:

1. Start the program.
2. Read the pattern and input list.
3. Compare the first element of the pattern with the first element of the input.
4. If they match, recursively compare the remaining elements.
5. If both lists become empty, the pattern matches successfully.
6. If any element does not match, return false.
7. Stop.

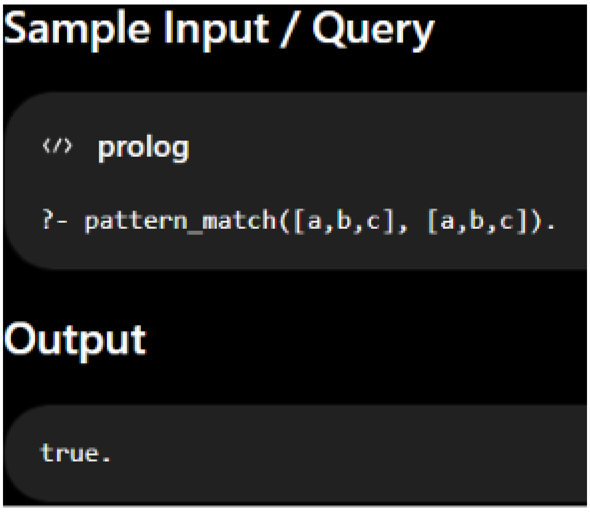
Prolog Program:

```
% Pattern Matching Program


pattern_match([], []).


pattern_match([H1|T1], [H2|T2]) :-
   H1 = H2,
   pattern_match(T1, T2)..
```

Output



Sample Input / Query

prolog

```
?- pattern_match([a,b,c], [a,b,c]).
```

Output

```
true.
```

Result:

Thus, the Prolog program successfully implements **pattern matching using unification and recursion**.